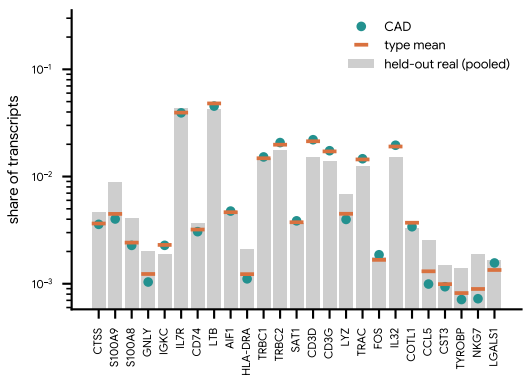
PBMC, conditional generation
unit: naive thymus-derived CD4-positive, alpha-beta T cell

```
Cell context: human peripheral blood
mononuclear cells (PBMC).
Cell type: naive thymus-derived CD4-positive,
alpha-beta T cell.
Gene expression frequency distribution of the
cell:
```

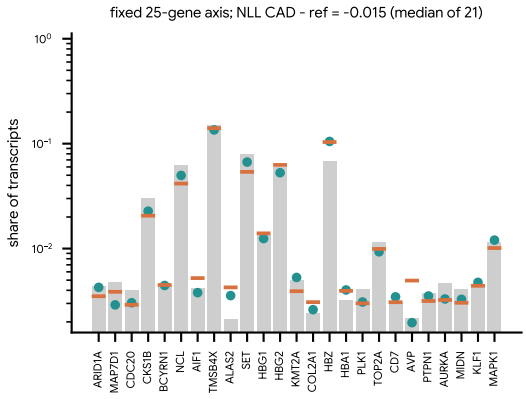
fixed 25-gene axis; NLL CAD - ref = +0.011 (median of 24)



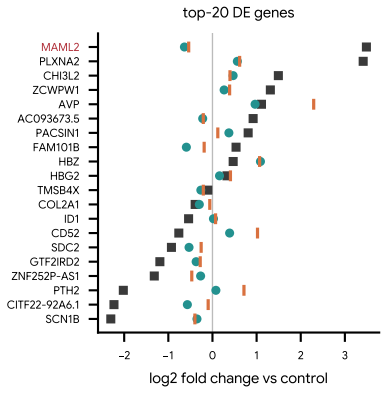
K562, no control cell, unseen single gene
unit: MAML2

```
Cell context: K562 cells, CRISPRa
perturbation.
Perturbation: MAML2.
Gene expression frequency distribution of the
perturbed cell:
```

fixed 25-gene axis; NLL CAD - ref = -0.015 (median of 21)



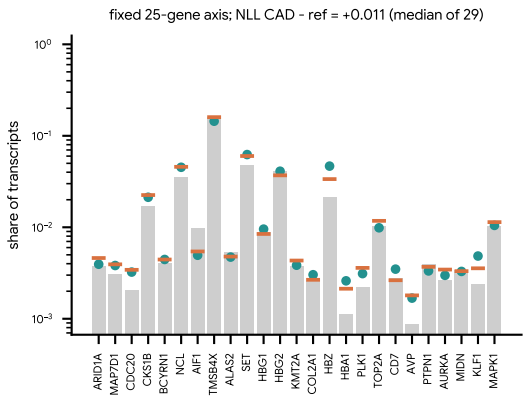
top-20 DE genes



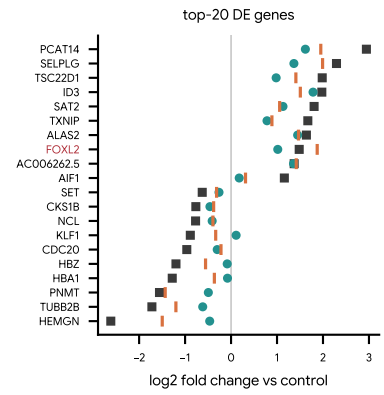
K562, no control cell, pair with one unseen gene
unit: FOXF1+FOXL2

```
Cell context: K562 cells, CRISPRa
perturbation.
Perturbation: FOXF1+FOXL2.
Gene expression frequency distribution of the
perturbed cell:
```

fixed 25-gene axis; NLL CAD - ref = +0.011 (median of 29)



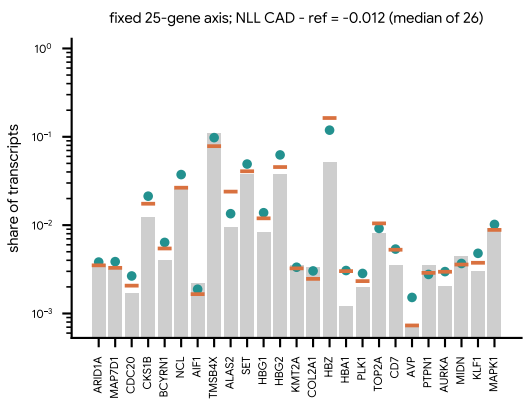
top-20 DE genes



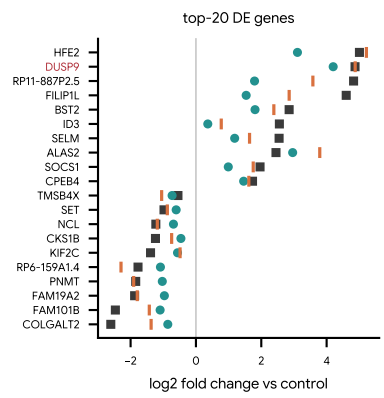
K562, no control cell, pair of two seen genes
unit: DUSP9+SNAI1

```
Cell context: K562 cells, CRISPRa
perturbation.
Perturbation: DUSP9+SNAI1.
Gene expression frequency distribution of the
perturbed cell:
```

fixed 25-gene axis; NLL CAD - ref = -0.012 (median of 26)



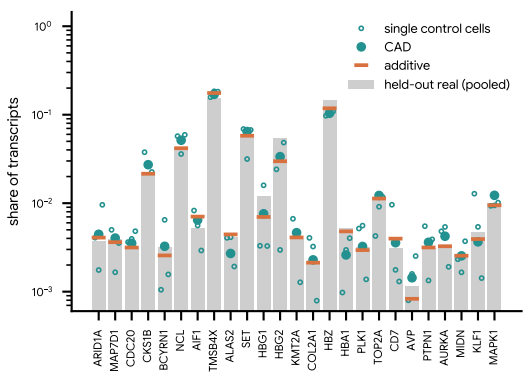
top-20 DE genes



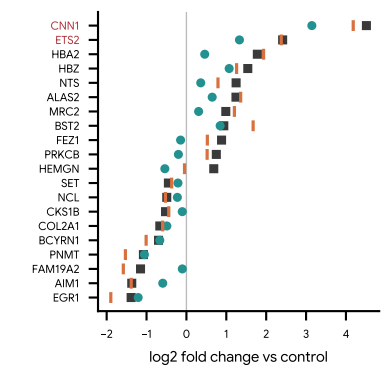
K562, control cell as text and residual, pair of two seen genes
unit: ETS2+CNN1

```
Cell context: K562 cells, CRISPRa
perturbation.
Perturbation: ETS2+CNN1.
Control cell (top genes with log-mass level):
TMSB4X:10 HBZ:9 SET:9 HBG2:8 CKS1B:8 NCL:8
KLF1:8 HBD:7 CD7:7 MAPK1:6 SMIM1:6 PLK1:6
MIDN:6 IL6ST:6 COL2A1:6 KMT2A:6 TSC22D1:5
ALAS2:5 SPI1:5 TAS2R14:5 ASCL2:5 RP6-159A1.4:5
HGSNAT:5 HIST2H2AC:5 ADSSL1:5 PNMT:5 MAP7D1:5
PIF1:5 ARID1A:5 AURKA:5 SAMD1:5 PTPN1:5
GS1-124K5.11:5 RP11-481J2.4:5 PLK4:5 GDPD1:4
TOP2A:4 BCAS3:4 PYY:4 NBR1:4 ACAP1:4
CTD-2095E4.5:4 COTL1:4 RP13-516M14.1:4
PMAIP1:4 IGFBP4:4 STIL:4 KIF18B:4
Gene expression frequency distribution of the
perturbed cell:
```

fixed 25-gene axis; NLL CAD - ref = +0.006 (median of 26)



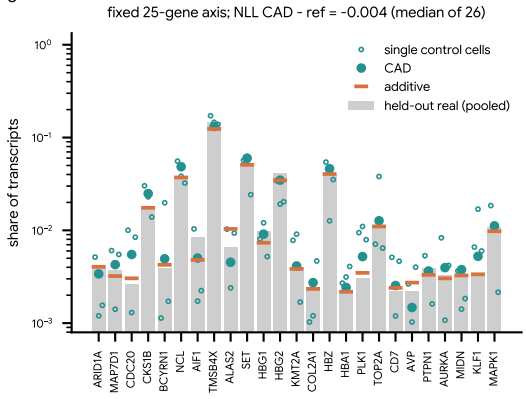
top-20 DE genes



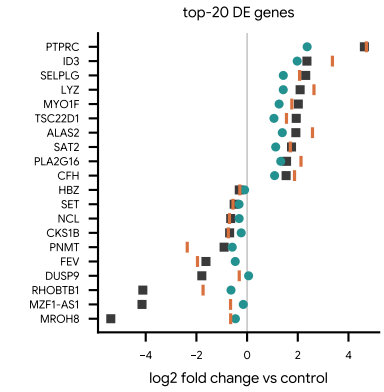
K562, control cell as residual only, pair of two seen genes
unit: FOXA1+FOXL2

```
Cell context: K562 cells, CRISPRa
perturbation.
Perturbation: FOXA1+FOXL2.
Gene expression frequency distribution of the
perturbed cell:
```

fixed 25-gene axis; NLL CAD - ref = -0.004 (median of 26)



top-20 DE genes



PBMC, zero-shot cell type (fold 0)
unit: naive thymus-derived CD8-positive, alpha-beta T cell

```
Cell context: human peripheral blood
mononuclear cells (PBMC).
Cell type: naive thymus-derived CD8-positive,
alpha-beta T cell.
Definition (Cell Ontology CL:0000900): A
CD8-positive, alpha-beta T cell that has not
experienced activation via antigen contact and
has the phenotype CD45RA-positive,
CCR7-positive and CD127-positive. This cell
type is also described as being CD25-negative,
CD62L-high and CD44-low.
Gene expression frequency distribution of the
cell:
```

fixed 25-gene axis; NLL CAD - ref = -0.728 (median of 4)

